

Andrew Andrusenko

Full-Stack / Frontend Engineer

Location: Ready to Relocate to the EU/Asia (Preferences: Germany, Cyprus, Spain)

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Professional Summary

Full-Stack Engineer with a solid background in FinTech architectures, specializing in **Angular, Node.js, and PostgreSQL** performance optimization. Strong track record of combining a software engineering skills with an extensive financial background to build production-grade, highly secure FinTech architectures. Significant experience in engineering high-frequency real-time systems, designing microservice infrastructures, and writing optimized database layers. Background in technical autonomy—moving from managing financial operations and driving "Shadow IT" automations in top-tier financial institutions to full-scale software architecture and product development. Fully mobile and seeking Frontend or Full-Stack roles within dynamic engineering teams.

Technical Skills

- **Frontend:** Angular (Standalone, RxJS, Material Design, Jest), Data Visualization (Konva, Elk, Dagre), Web Performance, Vanilla UI.
 - **Backend & Databases:** Node.js (Express, Passport.js), PostgreSQL (RLS, Custom Functions/Triggers), Redis, MongoDB, Python (Flask).
 - **Infrastructure & DevOps:** Docker, Docker Compose, Microservices, Network Isolation, Brotli Compression, Browser Automation.
 - **Architecture & Domain:** Real-Time Data (WebSockets), Scalable REST APIs (100+ Endpoints), Dynamic Query Pooling, JWT Auth Systems, Financial Accounting, Portfolio Returns (TWRR, FIFO, HWM calculations).
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Professional Experience

Full-Stack / Frontend Engineer | System Solutions Inc (Tech Vendor)

Feb 2025 – Present | Moscow, Russia

Product Context: A distributed, microservice-based low-code platform for automated business process setup, task management, and inter-domain contract execution across isolated corporate infrastructures.

[Live Demo](#) | [Live Demo 2](#) | Code Repository: Private (Available for live review during interviews)

[Live Real-Time Quotes WSS](#) | [Frontend Code Repository](#) | [Backend Code Repository](#)

- **Architected and developed an in-house data visualization library** using **Konva, Elk, and Dagre**, enabling interactive management of complex business process diagrams and contracts.
- **Led a major architectural refactoring** to integrate modern Angular best practices, migrating the codebase to **Standalone components**, implementing **@defer loading**, and leveraging **RxJS** for reactive state management.

- **Engineered a high-frequency real-time stock quote system** utilizing **WebSockets** with updates up to 50ms; optimized rendering performance via **RxJS buffering, throttling, and state caching** to prevent UI freezing.
 - **Drove a 6.5x reduction in bundle size**, breaking down a single 10MB chunk into 5.5MB of optimized small chunks, and further compressing it to **1.5MB via Brotli**, significantly improving initial page load time.
 - **Successfully assembled, updated, and containerized a complex Python (Flask) backend** comprising 6 microservices from fragmented source pieces to set up a fully functional, localized live staging environment.
 - **Optimized infrastructure footprint by cutting Docker image sizes from 1.44 GB to 300 MB per image** and established secure service communication across 3 distinct Docker networks/volumes.
 - **Proactively detected and resolved critical memory leaks** and rendering bottlenecks, ensuring long-session application stability and smooth UI performance.
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Principal Software Architect & Founder | Enterprise Asset Management Platform (AMP)

Mar 2022 – Sep 2024 | Remote / 3A Solutions

 [Live Demo](#) | Code Repository: Private (Available for live review during interviews)

Product Context: A full-scale Portfolio & Order Management System (PMS/OMS) for investment firms, automating the entire investment lifecycle from risk-profiling to client-side trade allocation.

- **Architected and engineered the comprehensive FinTech platform from scratch**, delivering a fully decoupled, scalable architecture to manage the full investment cycle and double-entry accounting.
 - **Designed and deployed a highly secure, scalable PostgreSQL architecture** comprising 69 tables, 100 constraints, 13 triggers, and **18 Row-Level Security (RLS) policies** to guarantee absolute multi-tenant data isolation.
 - **Developed 82 custom database functions and procedural scripts** to shift heavy financial math to the database layer, automating FIFO calculations, balance projections, and model-portfolio-based order execution under tight investment restrictions.
 - **Engineered an automated rebalancing and order routing engine** that overlays model portfolio structures onto active client accounts, calculates asset deviations against live market data, aggregates individual orders into bulk trade tickets, and executes multi-client allocations post-trade.
 - **Built a proprietary FinTech analytics engine** capable of calculating complex fee structures (including Management/Performance fees, High-Water Mark, and sliding scales) and analyzing portfolio performance via **Time-Weighted Rate of Return (TWRR)** and P&L factor analysis.
 - **Built a robust Node.js REST API layer with 100+ endpoints**, integrating secure session authentication via Passport.js and third-party market data API integrations.
 - **Implemented an advanced security pattern** that dynamically switches the database query pool for each incoming request, mapping application-level user roles directly to 6 database group roles.
 - **Created a high-performance, responsive Angular Frontend** utilizing Material Design and RxJS streams, optimizing complex financial dashboards, return-rate graphs, and multi-asset data visualization for high-velocity user workflows.
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Lead Software Architect | Central Authentication Service & Identity Provider

Sep 2024 – Dec 2024 | Remote / Proprietary Infrastructure R&D

[🔗 Live Demo](#) | [Backend Code](#) | [Frontend Code](#) Product Context: A decoupled, reusable central identity Service module engineered to manage secure sessions, user onboarding, and access control for cross-origin web ecosystems.

- **Architected and developed a standalone Central Authentication Service (IdP)** designed as a highly reusable infrastructure component for cross-origin enterprise applications.
- **Implemented a robust client-side security architecture** using an **HttpOnly, Secure, and SameSite=Lax cookie-encapsulated JWT strategy**, eliminating local storage token footprint to guarantee absolute protection against XSS vulnerabilities.
- **Engineered a non-blocking, lockless token synchronization engine** using the **RxJS exhaustMap Semaphore Pattern** inside global Angular Functional Interceptors (**HttpInterceptorFn**); effectively manages queued in-flight requests and prevents API token-refresh flooding during expired sessions.
- **Designed an automated transport-level handler** via an isolated **withCredentials** interceptor chain, dynamically modifying cross-origin CORS pipelines without leaking environment-specific configs into business logic.
- **Developed dynamic polymorphic subsystems**, including an interactive Admin Role Guard system and a single-view adaptive credential restoration module morphing states based on active route parameters and anti-spam cooldown mechanisms.

Back-Office Specialist & Operations Automator | BNP Paribas (Global Banking)

Mar 2021 – Jan 2023 | Moscow, Russia

- **Leveraged a lifelong passion for software development** to identify operational bottlenecks and independently build custom automation tools that optimized core banking workflows.
- **Engineered a custom BackOffice Dashboard** utilizing a Vanilla Web UI, Node.js, and SQL Server, delivering a flexible interface for real-time visualization and processing of global trades and payments.
- **Developed an automated tax suspension verification service** using browser automation and custom scripting, eliminating intensive manual verification cycles for clients applying for deposits.
- **Designed and implemented an internal trade confirmation tracking module**, creating an intuitive interface for real-time status management, document workflows, and automated transaction reconciliation.

Quantitative Developer & Private Investor | Own Asset Portfolio (Self-Employed)

Aug 2016 – Feb 2021 | Moscow, Russia Tech Stack: Python, Node.js, SQL, Excel/VBA, Financial Analytics

- **Designed and built proprietary algorithmic trading scripts and data parsers** to automate market data collection, technical analysis, and portfolio risk management.
 - **Developed custom analytical dashboards** (Node.js/SQL) to track asset allocations, evaluate historical performance metrics, and model multi-asset investment strategies.
 - **Successfully managed an independent multi-asset investment portfolio**, applying analysis, modeling, and risk-mitigation frameworks.
 - **Dedicated extensive depth to technical self-education**, mastering advanced database management (SQL), data structures, and architectural software patterns applied to financial systems.
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Head of Operations & Support / "Shadow IT" Dev | Renaissance Investment Management

Feb 2006 – Jan 2011 | Moscow, Russia

- **Managed the launch of 2 critical middle-office platforms** to support the consecutive growth of the private client asset management business.
 - **Independently coded and developed a crucial extension module** for the core middle-office platform under tight deadlines, preventing launch delays and enabling continuous iterative upgrades aligned with business demands.
 - **Translated complex business requirements into software logic**, designing, testing, and continuously improving the portfolio management system for traders and portfolio managers.
 - **Created automated workflows for investment declaration compliance**, implementing programmatic controls to monitor investment restrictions within clients' orders and strategies.
 - **Maintained the portfolio management application**, managed live data feeds, and built custom reporting tools for the active trading floor.
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Education & Self-Education

Degree in Finance (Graduated with Honors)

State University of Management / Central Bank of Russia Banking School

- **Bridge to Engineering:** My formal background in finance provides me with a deep domain understanding of FinTech architectures, complex accounting algorithms, and asset management models.

Software Engineering & Systems Development

Self-taught programmer with lifelong active, hands-on coding experience

- Continuous deep dive into Advanced Data Structures, System Design, Algorithms, and Software Architecture Patterns.
 - Built and maintained complex full-stack web production architectures from scratch.
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Personal & Languages

- **Languages:** English (C1 – Full professional proficiency) | German & Spanish (B2 – Reading proficiency) | Russian (Native).
- **Interests:** Road cycling, tennis and books.